

4. FEASIBILITY STUDY

4.1 Operational Feasibility

Operational feasibility looks at whether the proposed system, MediSync, can work well inside the daily operations of the National University Dasmariñas clinic, and whether the people who will use it are ready and willing to adopt it. This section checks if the system answers the real problems of the clinic, if it fits the way the staff already work, if the users can learn it without great difficulty, and if the clinic can keep using it after it is launched. The findings below are based on the interview with Nurse Kyle Guntang, the data summary prepared from that interview, the initial requirements of the project, and the blank forms currently used by the clinic.

4.1.1 Current Operation of the Clinic

The university clinic is the main health facility on campus and it is open to everyone. It serves college students, senior high school students who are minors, faculty, staff, employees, executives, and outside visitors. On an average day, about 100 people visit the clinic for medical clearances, standard check ups, dental check ups, emergencies, and requests for medicine that does not need a prescription. Since the clinic is open for 14 hours a day, this is around seven visits every hour on average, and the crowd becomes heavier during busy periods such as freshman enrollment.

The clinic is open from Monday to Saturday, 7:00 AM to 9:00 PM. Five nurses work in rotating shifts of 7:00 AM to 4:00 PM, 10:00 AM to 7:00 PM, and 12:00 PM to 9:00 PM. The clinic follows the standard of one nurse for every 3,000 students, so with around 9,000 students the five nurses are considered enough. The nurses do not change shifts during peak seasons, but they render overtime when more manpower is needed. One doctor is available from Monday to Friday, 9:00 AM to 6:00 PM, and a reliever doctor is added when many medical certificates are needed. The dentist is available on Tuesday, Wednesday, and Friday from 9:00 AM to 6:00 PM, although the dentist is sometimes absent on Fridays, so the clinic mainly counts on Tuesday and Wednesday. The clinic is also supported by 30 to 40 student volunteers who are not medical staff and who are now being organized into a Red Cross group.

The clinic has five beds, a dental room with a dental chair, a lactation room, an examination table, a receiving area, and a restroom for patients only. Wheelchairs and a stretcher are placed in the lobby. The clinic works only as a first aid station, so severe cases that need more than first aid are referred to a hospital. The standard of one bed for every 1,000 students is not yet met, and an order for one or two more beds has been placed.

Every patient follows the same procedure, whether the patient is a student, a faculty member, or a staff member. The clinic uses triage, which means the most urgent cases are served first. Accidents, bleeding, and fainting are handled right away, while cases such as cough, red eyes, and clearance

requests are delayed and may be asked to wait outside when the clinic is crowded. The nurse asks the purpose of the visit, gets the paper folder of the patient from the filing cabinet, takes the blood pressure, heart rate, and oxygen level, and hands the paper record to the doctor. After the consultation, the patient may receive medicine, rest in a bed for 30 minutes to 1 hour, go back to class, go home, or be referred to a hospital. Senior high school students who are minors are released only to a parent, while college students may go home on their own.

Records are kept in locked cabinets arranged alphabetically by academic program. Medical certificates and vital signs are written by hand. Medicine is counted by hand at the end of each day and typed into an Excel file, newly arrived medicines and supplies are written in a logbook, and the monthly report is made from the Excel file. The schedule of the doctor and dentist is posted on the Facebook page of the clinic, and appointments are handled through a Microsoft Forms page that does not show how many patients are waiting. The university has a system called NUIS, but the clinic uses it only in a limited way.

4.1.2 Operational Needs Addressed by the System

MediSync is built around the problems that the clinic staff described during the interview. Each main feature answers one of these needs, and each feature replaces a task that the staff already do by hand. This makes the change easier to accept and easier to carry out in daily work.

The first need is a reliable place for patient records. Paper folders are sometimes filed under the wrong program, and when a folder cannot be found, a new one is made, which creates two records for one student. The Electronic Health Records feature lets the nurse type patient information straight into the system and find a record by searching. The search through filing cabinets is no longer needed, and each patient keeps only one record. Nurse Kyle Guntang said that typing into a computer is faster than writing by hand, which shows that the staff already see the value of this change.

The second need is a visible and organized queue. At present, patients cannot see how many people are waiting or whether the doctor is around. The Smart Appointment Management feature lets a student book a visit for a non emergency concern, and the Digital Queue System shows the patients who are waiting. Before the visit, the student answers short questions about symptoms such as fever, headache, injury, and allergies, and the system places the case in a low, medium, or high priority queue. This matches the triage rule that the clinic already follows, because urgent cases are still served first. The difference is that the sorting starts before the patient arrives, so the lobby is less crowded and serious cases are easier to notice. When a patient with a booked appointment arrives, tapping their ID at the clinic entrance confirms their arrival and matches it to that appointment automatically, so the nurse at the front does not need to search for the booking by name.

The third need is a clear and shared schedule. The schedule of the doctor and dentist is posted only

on the Facebook page, and many students do not follow it, so some of them visit when the doctor is not around and then have to come back. MediSync shows the schedule of the doctor and dentist inside the appointment feature, so students can check it before going to the clinic and avoid wasted trips.

The fourth need is faster and more accurate handling of medicine. Counting medicine by hand at the end of each day takes time and can lead to mistakes. The Medicine Inventory Tracker keeps a live record of stock, sends a low stock alert, and creates the monthly consumption report by itself. The Prescription Management feature connects this to the consultation. When the doctor issues a prescription, the system deducts the medicine from stock, records the transaction, updates the history of the patient, and prints the prescription. The doctor and nurses no longer need to repeat the same update in the logbook, in the Excel file, and in the patient folder.

The fifth need is a consistent way of issuing medical certificates. Certificates are now written by hand, and the doctor records only what was found during the consultation. The Medical Certificate Generator creates a printable certificate directly from the consultation record. The doctor remains the one who decides what is written, so the current practice of not excusing a student who has no valid findings stays in place. Because the certificate comes from the recorded findings, the document matches the actual consultation, and there are fewer disagreements with students.

The sixth need is easy access to clinic data. The monthly report is now made by counting from the Excel file by hand. The Health Analytics Dashboard shows completed consultations, released clearances, and medicines given out, and it can build reports by group such as students, faculty, and staff. This gives the nurses the same information they need for the monthly report without daily manual input.

The seventh need is protection of patient information. Once records are digital, there is a risk that a person without approval can view or change them. The Role Based Access and Security Management feature limits what each type of user can see or change, so private information stays available only to the people who need it.

The eighth need is clear communication with students and, where needed, their parents. Right now, follow up messages depend on whoever happens to be free to call or text a student, with no shared record of what was said. The Communication Module lets clinic staff send a message about an appointment, a follow up, or another concern directly to a student or, for a minor, to a parent or guardian, and every such message is kept on file for reference. Notifications for appointments, queue updates, and clinic announcements go out both inside the app and through the university's existing Outlook email, so a student does not need to check a Facebook page to know their schedule changed.

The ninth need is a clear record of consent before a person's information is collected. The clinic currently has no formal step where a patient agrees to how their information will be handled. MediSync presents a terms and conditions and consent notice at registration or first login, and requires agreement before any information is stored, which supports the clinic's compliance with the Data Privacy Act of 2012 discussed further in section 4.1.8.

4.1.3 Fit with the Existing Workflow and Policies

MediSync changes how information is recorded and shared, but it does not change how the clinic gives care. The triage rule, the order of the visit from purpose to vital signs to consultation, the limit of 30 minutes to 1 hour for resting in a bed, the referral of severe cases to a hospital, and the rule that minors are released only to a parent all stay the same. The doctor and the nurses remain the ones who make medical decisions. The system supports them by giving faster access to information and by handling repeated paperwork. Because of this, the staff do not need to learn a new way of caring for patients. They only need to learn a new way of recording it.

The clinic also already uses standard numbered forms under the Health Services Office. These include the clinic visit log sheet, the incident report, the medical certificate, the prescription, the medical pass, the excuse letter, the go home form, the pre employment certificate, the health examination form, the medical records form, and the dentist referral and chart forms. MediSync is planned to copy the fields of these forms in digital pages, so the staff will see familiar layouts and will not be asked to record new kinds of information. Nurse Kyle Guntang agreed to provide blank copies of the forms so the team can follow the exact fields in use. Only blank copies are shared because of the Data Privacy Act.

The staff are also already used to digital tools. Nurse Kyle Guntang created the appointment form in Microsoft Forms, the medicine record is kept in an Excel file with formulas, and forms and messages can be sent through Microsoft Teams and Outlook. This shows that the staff have the basic computer skills needed to use a clinic system, and it also means the university's existing Outlook infrastructure can be reused for MediSync's notifications instead of the team having to build and maintain a separate messaging service.

MediSync also fits how the clinic already treats anyone who needs care, not only enrolled students. The clinic already serves faculty, staff, employees, and outside visitors the same way it serves students, so MediSync carries this over by giving employees the same booking and account access as students, while outside visitors who have no account are registered manually by clinic staff at the point of care, the same way they are handled today.

MediSync will be delivered through two connected parts, a mobile application that students,

employees, and visitors with an account use to book a visit, check their queue, view their own record, and access the health engagement feature, and a web based portal that clinic staff use on the clinic's computers. Since the mobile app is meant to run on a phone students already own, this does not require them to buy a new device, though they will need to install the free application, which is a smaller step than the account creation they already do for Microsoft Forms.

4.1.4 User Characteristics and Readiness

The system has several types of users, and each group is ready for MediSync in a different way.

The five nurses are the main users. They receive the patients, take the vital signs, search and update records, manage medicine stock, and prepare documents. They already work with Excel and Microsoft Forms every day, so they have the basic skills needed. What they need most is training on the new screens and steps.

The doctor and the dentist use the system during consultations to view patient history, write prescriptions, and create medical certificates. The dentist also uses the dental chart and the referral form. Consultation time is limited, so the screens for these users must be short and must need only a few steps. Since the system pulls information from the patient record, they do not need to write the same details again.

The student volunteers are non medical staff. Their access should stay limited to simple tasks such as helping with the queue on busy days, and they should not be able to open medical records or medicine stock. They are already required to keep information confidential through the volunteer agreement of the Health Services Office.

Students, including senior high school students who are minors, are the largest group of patients. They already use online forms, the university system, and social media, and most of them use a phone every day, so installing the mobile app, booking an appointment, checking the queue, tapping their ID at the clinic entrance, and using the health engagement feature do not need skills beyond what they already have. For minors, the rule that a parent must pick them up stays in place.

Faculty, staff, employees, executives, and outside visitors follow the same procedure as students, with employees able to book ahead and tap in on arrival the same way students do. Outside visitors and patients with emergencies may not have an account or the time to book, so the walk in option remains available, and a nurse can enter their details at the receiving area.

Clinic management and university administrators use the dashboard to view reports and plan medicine supply and services. They do not need special skills because the reports are shown in a simple form.

The interview shows that the clinic is open to the change. Nurse Kyle Guntang said that the clinic does not yet have a system for electronic health records. He also described what would be most helpful, which is a view of the queue, the number of consultations made, the clearances released, and the medicines released, together with faster extraction of reports. These match the features of MediSync. He also explained that moving from paper to electronic records is hard, and he described the loss of records as time consuming but not a very serious problem. This means the staff are used to the paper process and may not feel a strong push to change at first. For this reason, the system must show quick and clear gains in daily work, and the change must be supported with proper training.

4.1.5 Expected Operational Benefits

For the nurses, the biggest gain is less manual work. They no longer search filing cabinets, no longer refile folders, no longer count medicine at the end of each day and type the numbers into Excel, and no longer count consultations by hand for the monthly report. Vital signs are typed once into the record and can be seen by the doctor right away. The time saved can be used to attend to patients, which is important in a clinic that receives about 100 visits a day.

For the doctor, the history of the patient is ready at the start of the consultation. Prescriptions and medical certificates are made from the same record, and the updates to stock and history happen by themselves. This shortens the paperwork after each consultation and leaves more time for the patient.

For the dentist, the dental chart and referral form become part of the same digital record, so the dentist no longer depends on paper files that may be in the wrong place.

For students, the visit becomes more planned. They can check the schedule before going to the clinic, book a slot, and see the queue. This reduces wasted trips, long waiting in the lobby, and lost class time. Medical certificates and prescriptions are also released faster. Direct messages and Outlook notifications also mean a student finds out about a clinic announcement or a change to their appointment without needing to watch the clinic's Facebook page, and the health engagement feature gives them an easy way to pick up basic health information between visits.

For the clinic and the university, records become secure, easy to find, and free of duplicates. Staff have a live view of medicine stock, which helps prevent shortages, and reports are ready when needed. The clinic and the university gain reliable data on patient volume, common illnesses, and medicine use, which can guide future decisions.

The system does not add beds or staff, so the benefits are in the flow of work and not in the physical size of the clinic. Still, a smoother flow makes better use of the five beds and the five nurses that the

clinic already has.

4.1.6 Operation During Peak Periods

The clinic is hardest to manage during periods such as freshman enrollment, when many students need medical clearances at the same time. During these periods, the nurses render overtime and a reliever doctor is added, but the lobby still becomes crowded because everyone arrives without a schedule.

MediSync helps in four ways. First, the digital queue is built for a large number of students, so the clinic can handle many clearance requests in an organized manner. Second, appointments spread the arrival of students across the day instead of letting everyone come at once. Third, the priority queue keeps urgent cases from waiting behind routine ones. Fourth, tapping an ID at the door confirms arrival in seconds instead of a nurse searching for the student's name, which matters most exactly when the lobby is fullest. The clinic already treats clearance requests as cases that can wait, and the queue lets these students wait outside the clinic until their turn is near instead of standing in the lobby. The Medical Certificate Generator also speeds up the release of documents, which reduces the load on the reliever doctor.

The system cannot remove the need for overtime or a reliever doctor during the busiest weeks, but it lowers the crowd and the stress on the staff. This makes the current staffing more manageable.

4.1.7 Training, Support, and Transition

A new system is only useful when the people can use it with confidence, so a clear training and transition plan is part of operational feasibility.

The nurses, the doctor, and the dentist should attend a short hands on orientation before the system starts. The training should use sample patients and the real forms of the clinic, so the staff practice the same steps they do every day, such as checking in a patient by tapping an ID or searching manually, entering vital signs, issuing a prescription, sending a message to a student or parent, and creating a medical certificate. A short guide with pictures should be given to each user, and the Help feature of the system can be used for quick questions.

The student volunteers should receive a shorter orientation that covers only their limited tasks and the rules on confidentiality. Students should be informed through the channels that the clinic already uses, such as the Facebook page and Outlook messages, together with posters near the clinic, and should be walked through installing the mobile app as part of onboarding. The student pages should be simple enough that no long training is needed.

The change from paper to digital should be done in steps. For a short period, the clinic can use both

paper and the system together, so the staff are not left without a backup while they are learning. Records of students who visit the clinic should be entered first, and older folders can be entered over time. While entering old records, duplicate folders of the same student should be merged into one profile. Once the staff are comfortable and the data is complete, the clinic can slowly stop using the paper folders.

During the first weeks, one nurse can serve as the main contact for questions and problems, and the project team should be available to fix errors and take feedback from the staff. This makes sure the system is adjusted to the real needs of the clinic before it becomes the main way of working.

4.1.8 Data Privacy and Security in Daily Operation

The clinic handles sensitive personal information such as health records, and some of it belongs to minors. Because of this, the system must follow the Data Privacy Act of 2012. The clinic already shows care for privacy. Records are kept in locked cabinets, only blank forms are shared with outside groups, and volunteers sign a confidentiality agreement before they serve.

MediSync continues this practice in digital form. Every user has a separate account, and access depends on the role of the user. Students can see only their own appointments and records. Nurses can view and update patient records and medicine stock. The doctor can view consultation records and issue prescriptions and certificates. The dentist can view and update dental records. Volunteers have very limited access. Administrators view reports without needing to open the full details of a patient. In this way, the locked cabinet is replaced by accounts that control who can reach what.

Before a student, employee, or visitor's information is entered into MediSync for the first time, the system records their agreement to a terms and conditions and consent notice explaining how their information will be used, giving the clinic a documented step that the current paper process does not have.

The system should also keep a record of who opened or changed information, so any misuse can be traced. During the development and testing of the system, only sample data should be used, and no real patient records should be entered until the system is ready and the proper approval is given.

4.1.9 Operational Risks and Ways to Reduce Them

The first risk is that some staff may resist the change because they are used to paper. This can be reduced through early involvement of the nurses, training that uses their own forms, a period where paper and the system are used together, and quick fixes when they report problems.

The second risk is loss of internet or power. The clinic must still serve patients even when the system is down. The clinic should keep printed copies of its key forms, such as the visit log sheet, the

prescription, and the medical certificate, to use during a short interruption. After the system returns, the information written on paper should be entered so the digital record is complete.

The third risk is that some students may give untrue answers in the symptom questions to move ahead in the queue. To reduce this, the nurse should be able to review and change the priority level when the vital signs or the appearance of the patient do not match the answers. In this way, the clinical judgment of the nurse stays above the automatic sorting.

The fourth risk is that some patients cannot or will not use the online booking. This includes outside visitors and patients with emergencies. The walk in option stays available and urgent cases are still served first.

The fifth risk is data entry mistakes. Some fields can be made required and limited to valid choices, and the doctor and nurses can review the entries during the consultation.

The sixth risk is unauthorized access to records. This is reduced through separate accounts, role based access, strong passwords, and the record of who accessed the information. Staff and volunteers must also continue to follow the confidentiality rules of the clinic.

The seventh risk is that the number of computers or devices at the clinic may not be enough for the nurses and the doctor to use the system at the same time. This is discussed further in the Technical Feasibility section, but it should be checked before the system starts.

The eighth risk is that the ID reader at the clinic entrance could stop working or a patient's ID could fail to register. If this happens, the nurse should be able to check a patient in manually by searching their name or student number, so a hardware problem does not stop the queue from working.

The ninth risk is that some students may not install the mobile app right away. Since booking and viewing records depend on the app, the clinic should keep the walk in option available for these students during the transition period, the same way it does for outside visitors, so no one is turned away for not having the app yet.

4.1.10 Organizational Support and Long Term Use

The clinic has shown support for the project. Nurse Kyle Guntang agreed to be interviewed, explained the operations and problems in detail, and agreed to provide blank copies of the forms. The clinic has also said that it has no electronic health record system yet, and it already tried to solve part of the queue problem with Microsoft Forms. This shows a real need and an interest in a better solution. The university already has tools such as Microsoft Teams, Outlook, and NUIS, which can support communication and later connection with other university systems.

For long term use, the daily work is simple. The nurses update medicine lists and the doctor and dentist schedule, and the administrators manage user accounts. No large technical team is needed for daily use. Still, the system needs someone to maintain it, update it, and fix problems after the project ends. The clinic and the university should decide who will take this role, such as the university IT office, before the system is fully launched.

4.1.11 Summary of Operational Feasibility

Based on the interview and the review of the current clinic process, MediSync is operationally feasible. It answers the real problems of the clinic, which are lost and duplicate records, long waiting and crowding, unclear schedules, manual medicine counting, handwritten certificates, slow reports, scattered communication, and the absence of a formal consent process. It keeps the medical rules and steps of the clinic and changes only how information is recorded. It copies the forms that the staff already use, and it works on devices that students already have. The users are open to the change, and the risks that were found can be handled through training, a step by step transition, backup forms, careful access control, and support from the clinic and the university. With these in place, the clinic can expect a more organized, faster, and safer daily operation.

4.2 Technical Feasibility

Technical feasibility looks at whether MediSync can actually be built and run with the hardware, software, and skills realistically available to the team and to the university clinic. This section compares what the clinic already has against what the system needs, and checks whether the intended users can handle the technical side of the system.

A. Hardware and Software Requirements vs Existing Setup

The university clinic currently runs on manual and paper based methods, with only a small number of shared computers used for typing reports and for opening the Microsoft Form that collects appointment requests. There is no dedicated server, no central database, and no software built to handle patient records, queueing, inventory, or a machine learning based triage feature. To implement MediSync and support close to a hundred patients a day, the clinic needs the hardware and software specifications below. Table 4.1 lists the hardware requirements, and Table 4.2 lists the software requirements.

Table 4.1. Hardware Specifications

Hardware	Existing	Required
Server	None	A dedicated application server or a cloud hosted equivalent with at least an Intel Core i5 processor
Client Workstations	A few shared desktop units used only for the Microsoft Form and Excel	At least five workstations, one each for the nurses station, the doctor, the dentist, the admin office, and the student volunteer counter, with an Intel Core i5 processor or higher
ID Tap Reader	None	An RFID or NFC reader installed at the clinic entrance to confirm a patient's arrival and match it to a booked appointment, with a second unit recommended for a busy secondary entry point
Internet Connection	Shared office WiFi with no fixed bandwidth	A stable connection within the university local area network with enough bandwidth to support several users at once
RAM	4GB or lower on the shared units	8GB or higher on each workstation and 16GB or higher on the server
Storage	Local hard drives with no backup routine	A 500GB solid state drive on each workstation and at least 1TB of storage with a backup drive on the server
Printer	A shared office printer	A dedicated printer for medical certificates, prescriptions, and monthly reports
Backup Power Supply	None	A backup power unit for the server to prevent data loss during power outages
Mobile Line or SMS Gateway	None	A registered SIM based line or an SMS gateway account for appointment and queue notifications

Table 4.2. Software Specifications

Software	Existing	Required
Operating System	Windows 10 or an older version on the shared clinic computers	Windows 11 or a supported Linux distribution for the server, with clinic computers updated to a supported version
Web Browser	Google Chrome, used mainly to open the Microsoft Form	Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari, kept on a current version
Database Management System	A single Excel file kept on one computer	A relational database system such as MySQL or PostgreSQL
Backend Framework	None	A Python based framework such as Django or Flask

Mobile Application Framework	None	A cross platform framework such as Flutter or React Native, so the patient facing app runs on both Android and iOS from one codebase
Machine Learning Library	None	A Python based machine learning library for training and running the triage classification model
Notification and Email Integration	None	Microsoft Graph API access under the university's existing Microsoft 365 subscription, used for in app and Outlook email notifications without a separate paid mail service
Document and Certificate Tools	Handwritten forms and printed templates	A PDF generation tool for building and signing digital medical certificates
Security Tools	None	Encryption and hashing tools to protect passwords and patient records

B. Availability of Required Hardware and Software

Most of the items listed above are not yet available at the university clinic. The clinic works with a small number of shared computers that were only ever meant for typing reports and opening the Microsoft Form, so they cannot support a live database, a real time queue, and a machine learning model running together, and the clinic has no ID reader of any kind at its entrance today. Before MediSync can be deployed, the clinic will need at least one server unit or a cloud hosted equivalent, upgraded workstations for each user group, one or two ID tap readers, and a stable connection within the university network that can support many users at once, especially during busy periods such as freshman enrollment.

On the software side, the operating system and web browsers already in use at the clinic can stay in place with minor updates, since the staff portal is meant to run inside a standard browser. What still needs to be added are the database system, the backend framework, the mobile application framework, the machine learning library for the triage feature, and an SMS gateway subscription for sending notifications. Because the project has a small budget, the team plans to rely on free and open source tools such as MySQL, PostgreSQL, Django, Flask, and Flutter, since these carry no licensing costs and are supported by enough tutorials and documentation for the student developers to learn during building and testing. The university's existing Microsoft 365 subscription similarly means Outlook based notifications add development time but no new licensing cost.

C. User Requirements vs User Profile

The system is designed for seven main user groups inside the university clinic. Table 4.3

compares what each group is already expected to know against the skills they will need once MediSync is in place.

Table 4.3. User Requirements vs User Profile

User Type	Existing Knowledge	Required Knowledge or Skills
Student	Basic digital literacy, comfortable using a smartphone or a web browser, familiar with online forms such as Microsoft Forms	Must be able to create an account, install and use the mobile app, book an appointment through the doctor and dentist schedule, answer the symptom questions, tap their ID to check in, check the queue number, view the parts of their health record that are open to them, and use the health engagement feature
Employee or Visitor	Basic digital literacy for employees with a university account, little or no prior exposure to the system for outside visitors	Employees must be able to use the mobile app to book an appointment and check in the same way students do, while outside visitors need no account since clinic staff enter their details manually at the point of care
Clinic Nurse	Healthcare training, familiar with daily clinic tasks, comfortable typing data into Word or Excel	Must be able to open a patient medical file, record vital signs on the system, manage the digital queue including the priority queue, confirm a patient's arrival through the ID tap or manually, update medicine stock counts, and send a message to a student or parent when needed
University Physician	Licensed physician, comfortable using simple web forms on a computer	Must be able to view a patient's full health history, write checkup notes, issue a digital prescription, and generate and sign a medical certificate
University Dentist	Licensed dentist, comfortable using basic computer software	Must be able to view dental histories, record treatment notes, and manage the dental appointment schedule
Student Volunteer	Friendly, good communication skills, understands basic clinic rules	Must be able to view public queue numbers and the daily staff schedule, while staying restricted from private student medical files
System Administrator	Background in computer science, knows database management and system security	Must be able to create and manage user accounts, assign roles, track system activity, perform database backups, and fix technical errors

4.3 Economic Feasibility

Implementing MediSync entails an initial investment in system development, hardware

acquisition, server and network setup, and user training. While the estimated figures below are assumption based, they reflect realistic market values for the technologies the system depends on, including a Python based web portal, a Flutter based mobile application, a relational database, an SMS gateway for notifications, ID tap readers, and a Python based machine learning library for the triage feature. Despite the upfront costs, the projected benefits in reduced administrative workload, faster record retrieval, and fewer manual errors make the system economically feasible for the university clinic of National University Dasmariñas over its useful life.

4.3.1 Economic Benefits, Estimated Annual Savings

1. Reduced Manual Record Search and Filing Time

Digitizing patient records removes the need for nurses to search through locked filing cabinets for every visit and prevents the creation of duplicate records when a folder is misplaced, freeing up time nurses can spend on direct patient care.

Estimated Savings: ₱42,000 per year, based on roughly one hour of search and filing time saved across the five nurses each working day.

2. Reduced Paper and Printing Costs

The shift from handwritten forms, prescriptions, and certificates to digital records lowers the clinic's ongoing spending on paper, ink, and filing supplies.

Estimated Savings: ₱2,400 per year.

3. Reduced Peak Period Waiting and Congestion Costs

Through the digital queue and appointment booking feature, staff spend less time managing an unorganized walk in line during peak periods such as freshman enrollment.

Estimated Savings: ₱18,000 per year.

4. Reduced Manual Inventory Counting Time

The Medicine Inventory Tracker Module automates the daily counting and logging of medicine stock that the five nurses currently perform by hand, freeing up time for patient care and lowering the chance of costly stock errors.

Estimated Savings: ₱30,000 per year.

5. Faster Prescription and Certificate Processing

The Automated Prescription Module and the Medical Certificate Generator Module let the doctor and dentist issue these documents directly from consultation notes instead of writing them by hand.

Estimated Savings: ₱4,800 per year.

6. Reduced Manual Communication Time

The Communication Module lets clinic staff send a reminder, a follow up, or an announcement through the app and Outlook instead of calling or texting each student individually, and automatic notifications remove the need for staff to send routine reminders by hand.

Estimated Savings: ₱6,000 per year.

7. Improved Monthly Reporting and Accountability

The Health Analytics Dashboard Module automatically compiles patient volume, common illnesses, and medicine usage into the monthly summary report, removing the need to recount handwritten records at the end of every month. Although not directly generating monetary savings, this contributes to more reliable data for administrators and stronger accountability across the clinic.

Total Estimated Annual Savings: ₱103,200

One Time Implementation Costs, Estimated

The one time cost of the MediSync system consists of System Development for the web portal, Mobile Application Development, Machine Learning Model Development, Hardware and Equipment, Server and Network Setup, SMS Gateway Integration, and User Training. The System Development cost of ₱150,000 covers the overall design and creation of the web portal used by clinic staff, including the database integration, the electronic health records module, the appointment and queue modules, the inventory and prescription modules, the communication module, and the interfaces built for each staff role. The Mobile Application Development cost of ₱80,000 covers the patient facing app used by students and employees, including account creation, appointment booking, the digital queue view, the health engagement feature, and the tap in check in function, built using a cross platform framework to keep the cost lower than developing separate native apps.

The Machine Learning Model Development cost of ₱15,000 covers the preparation of training data and the development and testing of the triage classification model used to sort symptom reports into low, medium, and high priority. The Hardware and Equipment cost of ₱188,000 covers the purchase of the server, five workstations, one or two ID tap readers, a dedicated printer, and a backup power supply, in line with the specifications outlined under Technical Feasibility.

The Server and Network Setup cost of ₱10,000 covers the configuration of the database, the application server, and the local network connection needed to run MediSync reliably inside the university clinic network. The SMS Gateway Integration cost of ₱5,000 covers the initial account setup and testing needed to enable appointment and queue notifications. Lastly, User Training and Orientation of ₱5,000 is allocated for orientation sessions with the nurses, the doctor, the dentist, the student volunteers, and the system administrator, covering the new screens, the ID tap check in

process, and the messaging feature.

Total One Time Implementation Cost: ₱453,000

Table 4.4. One Time Implementation Costs

System Cost Particulars	Cost in Pesos
System Development, Web Portal	₱150,000.00
Mobile Application Development	₱80,000.00
Machine Learning Model Development	₱15,000.00
Hardware and Equipment	₱188,000.00
Server and Network Setup	₱10,000.00
SMS Gateway Integration	₱5,000.00
User Training and Orientation	₱5,000.00
TOTAL	₱453,000.00

Recurring Costs, Estimated Annual

For the continuous operation and support of MediSync, annual recurring costs will be required. The Server and Network Maintenance cost of ₱8,000 per year covers the periodic upkeep of the local server, the network connection, and the database within the university clinic. The SMS Gateway Usage cost of ₱9,000 per year estimates the ongoing expenses for sending appointment confirmations, schedule changes, and queue notifications to registered students and employees.

In addition, the Database and Software Maintenance cost of ₱4,000 per year covers routine upkeep of the relational database and both the web portal and mobile app codebases, while the System Updates and Security Patches cost of ₱4,000 per year is allocated for periodic improvements and security enhancements that protect sensitive patient data across both platforms. Lastly, the System Maintenance and Repairs cost of ₱5,000 per year ensures preventive maintenance, performance checks, and minor repairs across the server, the five workstations, the ID tap readers, the printer, and the backup power supply.

Total Estimated Annual Recurring Cost: ₱30,000

Table 4.5. Recurring Costs

Recurring Cost Particulars	Cost in Pesos
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Server and Network Maintenance	₱8,000.00
SMS Gateway Usage	₱9,000.00
Database and Software Maintenance	₱4,000.00
System Updates and Security Patches	₱4,000.00
System Maintenance and Repairs	₱5,000.00
TOTAL	₱30,000.00

Taken together, the estimated annual savings of ₱103,200 cover the ₱30,000 in yearly recurring costs, leaving ₱73,200 each year. Against the one time implementation cost of ₱453,000, this places the simple payback period at just over six years. This is a long payback for a Software Engineering 1 prototype, and it reflects the fact that a clinic serving close to a hundred patients a day, now also supporting a mobile app and RFID hardware, is a genuinely larger undertaking than a simple form or tracker. The peso figure also does not capture benefits that are harder to price directly, such as fewer lost records, faster release of documents students need for class, and more reliable data for the university's own planning. For these reasons, MediSync is considered economically feasible over its useful life even though the payback period is not short, and the team recommends treating the mobile application and the mid range hardware as the biggest cost drivers to revisit first if the budget needs to be trimmed.

4.4 Schedule Feasibility

MediSync needs to be finished within the time given for Software Engineering 1 and the term that follows. This section checks if the planned work can fit that time. The schedule is based on the

team's Gantt chart, which covers five months and 20 weeks. Month 1 is September 2026, and the proposal defense is set for October 15 to 21, 2026. The dates should be adjusted to match the adviser's final approved calendar.

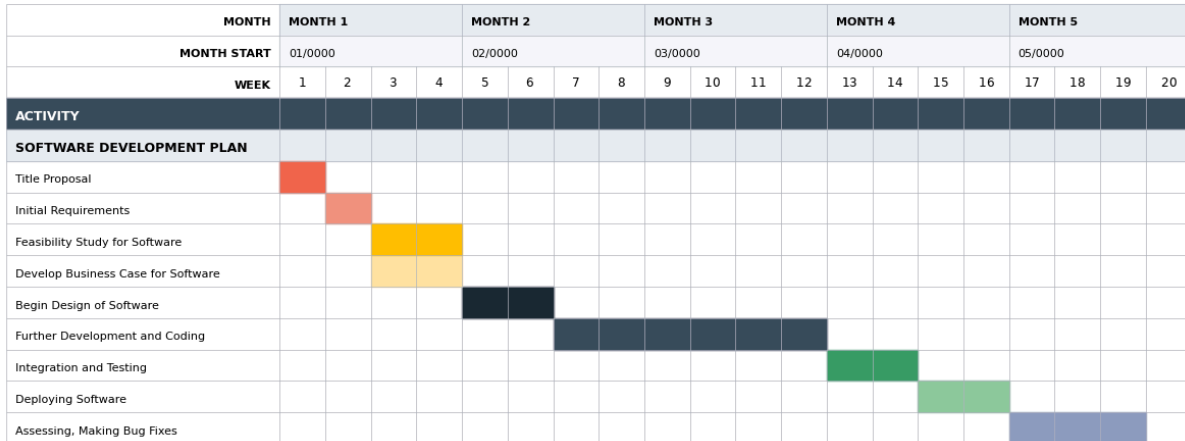


Fig 4.4.1 Gantt Chart - Software Development Plan

The Gantt chart shows nine activities in the order they will be done. The early activities are short because the team already finished most of the groundwork. The Title Proposal takes one week and the Initial Requirements take one week. The Feasibility Study and the Business Case for the software are done together in Weeks 3 and 4. Design begins in Weeks 5 and 6. The longest activity is Further Development and Coding, which runs for six weeks, from Week 7 to Week 12. It is followed by Integration and Testing, Deploying Software, and Assessing and Making Bug Fixes, which take the project to the end of the fifth month.

Table 4.6. Project Schedule

Phase	Gantt Weeks	Target Period	Key Activities
Title Proposal	Week 1	September 1 to 7, 2026	Title, problem statement, and initial scope approved
Initial Requirements	Week 2	September 8 to 14, 2026	Interview with the clinic nurse, data summary, and listing of the initial requirements
Feasibility Study and Business Case	Weeks 3 to 4	September 15 to 30, 2026	Operational, technical, economic, and schedule feasibility, together with the costs and savings of the system
Begin Design of Software	Weeks 5 to 6	October 1 to 14, 2026	Use case diagram, context diagram, database schema, and screen layouts
Proposal Defense and Panel Revision	Weeks 7 to 8	October 15 to 31, 2026	Formal presentation to the panel, followed by updating the documents based on panel feedback

Further Development and Coding	Weeks 7 to 12	October 15 to November 30, 2026	Setup of the project and database during the defense weeks, then building the priority modules in November, such as role based access, electronic health records, appointment and triage, digital queue, and medicine inventory
Integration and Testing	Weeks 13 to 14	December 1 to 14, 2026	Joining the modules, internal testing, and final submission of the Software Engineering 1 deliverables
Deploying Software	Weeks 15 to 16	December 15 to 31, 2026	Preparing the system for use in the clinic, which continues in the following term
Assessing and Making Bug Fixes	Weeks 17 to 19	January 1 to 21, 2027	Collecting feedback from users, fixing errors, and improving the system